Metformin is an oral medication widely used to manage type 2 diabetes mellitus. It belongs to a class of drugs called biguanides. It is considered a first-line treatment due to its effectiveness, safety profile, and low cost.

Mechanism of Action

Metformin works primarily by reducing glucose production in the liver (hepatic gluconeogenesis). It also improves insulin sensitivity in peripheral tissues such as muscles and fat cells.

Key actions:

- Decreases hepatic glucose output
- Increases insulin sensitivity
- Enhances peripheral glucose uptake
- Reduces intestinal glucose absorption

Metformin does not stimulate insulin secretion, which means it has a low risk of causing hypoglycemia.

Indications

Metformin is prescribed for:

- Type 2 diabetes mellitus
- Prediabetes (to delay progression to diabetes)
- Polycystic ovary syndrome (PCOS) (off-label use)
- Gestational diabetes (in some cases)

Dosage and Administration

Metformin is available in immediate-release (IR) and extended-release (XR) forms.

Typical dosage:

- Starting dose: 500 mg once or twice daily
- Maintenance dose: 1500–2000 mg per day
- Maximum dose: 2550 mg (IR), 2000 mg (XR)

It is usually taken with meals to reduce gastrointestinal side effects.

Pharmacokinetics

- Absorption: Moderate oral absorption
- Bioavailability: Approximately 50–60%
- Distribution: Not bound to plasma proteins
- Metabolism: Not metabolized in the liver
- Excretion: Eliminated unchanged via kidneys

Half-life: Approximately 4–9 hours

Side Effects

Common side effects:

- Nausea
- Diarrhea
- Abdominal discomfort
- Loss of appetite

Serious but rare side effect:

- Lactic acidosis (life-threatening condition)

Long-term use may lead to vitamin B12 deficiency.

Contraindications

Metformin should not be used in patients with:

- Severe renal impairment
- Acute or chronic metabolic acidosis
- Severe liver disease
- Conditions causing hypoxia (e.g., heart failure, respiratory failure)

Drug Interactions

Metformin may interact with:

- Alcohol (increases risk of lactic acidosis)
- Contrast agents used in imaging procedures
- Certain diuretics and antihypertensive drugs

Advantages

- Low risk of hypoglycemia
- Promotes mild weight loss or weight neutrality
- Improves lipid profile
- Cardiovascular benefits

Monitoring Parameters

Patients on metformin should be monitored for:

- Blood glucose levels
- HbA1c levels
- Kidney function (serum creatinine, eGFR)
- Vitamin B12 levels (long-term use)

Special Populations

Elderly:

- Dose adjustment required due to reduced kidney function

Pregnancy:

- Sometimes used but insulin is preferred

Children:

- Approved for use in children above 10 years